
Can Phase-Conditioned Imitation Recover Biomechanically Realistic Gait? A Walker2D Case Study.

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Abstract

Understanding human walking at the level of joint angles, ground reaction forces, and inter-limb timing matters for clinical biomechanics, rehabilitation, and assistive device design. Reinforcement-learning agents that imitate recorded human motion offer a tempting cheaper substitute for instrumented motion-capture labs: train a simulated body to track a reference, and the internal forces and gait-cycle structure should fall out of the physics for free. We ask whether that promise actually holds for a common research configuration. Can phase-conditioned PPO with a DeepMimic-style engineered reward, applied to MuJoCo Walker2d-v4 conditioned on real human inverse-kinematics reference data, recover a gait that holds up against held-out force-plate ground truth from the same subject? We train phase-aware PPO with a behavioral-cloning warm-start and reference-state initialization, alongside a parallel attempt at Adversarial Motion Priors (AMP). The phase-aware policies visually walk, but a held-out biomechanical-realism scorecard against the same subject's measured kinematics and ground reaction forces tells a different story. None of our trained variants pass. Peak vertical ground reaction force lands at 3.7 to 4.8 times body weight against a reference of 1.1, and double-support fraction sits below 2% against a reference of 23%. Those are the kinetic signatures of bouncing or running, not walking. The negative result motivates two paths forward we did not get to: AMP at GPU-parallel scale, where the discriminator escapes its small-data memorization regime, or a return to muscle-actuated models, where physiological force limits would make the running-gait failure mode structurally unreachable.

1 Introduction

Understanding human walking at the level of muscle forces, joint contact loads, and ground reaction forces matters for clinical biomechanics, rehabilitation, and assistive device design. The usual instrumented laboratory — optical mocap, force plates, EMG — costs upwards of \$150,000. Reinforcement-learning agents that imitate recorded human motion are a tempting alternative: if a simulated body tracks a reference closely enough, internal forces and gait-cycle structure should emerge from the physics for free.

From the original proposal to a scaled-down question. This project began as a study of *lab-to-field transfer in musculoskeletal reinforcement learning*: 3D, 80-muscle MyoLeg agents trained on lab-grade marker/force-plate/EMG references versus markerless smartphone mocap from OpenCap [14]. That scope was infeasible for one semester. We carried the spirit of the question into a smaller setting: the 2D MuJoCo Walker2d-v4 planar biped [12] with a single-subject IK reference at 1.25 m/s.

This paper. We train phase-conditioned PPO [10, 8] with a multi-term DeepMimic [6] reward and a behavioral-cloning warm-start, then evaluate the visually walking policies against held-out biomechanical ground truth from the same subject’s force-plate and IK data. We additionally implement Adversarial Motion Priors (AMP) [7, 2] as a learned-reward comparison. The visually walking policies produce running-gait kinetics; AMP at small environment counts collapses to a near-zero style reward, consistent with the GPU-parallel scaling the original paper required.

2 Related Work

DeepMimic [6]. The modern recipe for physics-based motion imitation: a phase-indexed reference drives a dense reward of weighted exponential penalties on pose, velocity, end-effector, and root tracking error, with Reference State Initialization for uniform phase coverage. Our reward and RSI follow this framework on a planar biped; unlike DeepMimic, we evaluate against held-out kinetic ground truth at test time.

Adversarial Motion Priors [7, 2]. AMP replaces hand-crafted imitation terms with a learned LSGAN discriminator over reference vs policy state-transitions plus a small task reward. Escontrela et al. [2] trained at 4096 parallel environments on a GPU cluster; we reuse their design at 8–16 CPU environments and reproduce the small-scale collapse implicit in their parallelism requirement. GAIL and AIRL [5, 3] are the discriminator-as-reward predecessors and share the same collapse pathology; we implement AIRL alongside AMP for comparison.

Massively parallel deep RL [9]. Rudin et al. cut legged-locomotion training time from hours to minutes with thousands of GPU-parallel environments — the empirical anchor behind the AMP scaling requirement and our planned MuJoCo MJX [4] port.

Muscle-driven imitation [11, 1]. KINESIS and KinTwin are recent imitation pipelines on Hill-type muscle humanoids: emergent activations correlate with recorded EMG without explicit EMG terms, and the approach reproduces able-bodied and impaired gait from markerless input. They motivate the muscle-actuation direction in Sec. 7, where physiological force limits make our Walker2d failure mode structurally unreachable.

3 Problem Statement

We formalize gait imitation as a Markov Decision Process $M = (\mathcal{S}, \mathcal{A}, T, r, \gamma)$ augmented with a reference motion. Let $q, \dot{q}$ be Walker2d’s joint positions and velocities, $s_{\text{base}} = [q, \dot{q}]$. The reference is a periodic sequence $R = \{q_t^{\text{ref}}\}_{t=0}^{T-1}$ of bilateral hip, knee, and ankle angles at 1.25 m/s, sampled at the simulator control rate. A scalar phase $\varphi \in \{0, \dots, T-1\}$ advances by one per step modulo T ; the phase-conditioned observation is

$$s = [s_{\text{base}} \mid q_{\varphi}^{\text{ref}} \mid \sin(2\pi\varphi/T) \mid \cos(2\pi\varphi/T)]. \quad (1)$$

The agent maximizes $\mathbb{E}_{\tau \sim \pi} [\sum_t \gamma^t r(s_t, a_t, \varphi_t)]$ where r encodes fidelity to R .

Held-out biomechanical evaluation. Separately, let B^* be biomechanical metrics from the same subject’s force-plate and IK data: stride period, cadence, double-support fraction, peak vertical ground reaction force (vGRF), and per-joint range of motion. The realism gap is the deviation $|B(\pi) - B^*|$, aggregated as a 0–4 progress score with a $\pm 20\%$ credible band per metric (Sec. 4.3). The training reward sees only R ; the kinetic targets in B^* are held out.

4 Method

4.1 Phase-conditioned PPO + DeepMimic reward

The Walker2d-v4 default observation is phase-blind. The policy sees joint angles and velocities but no information about where it is in the gait cycle. We augment the observation with the reference pose at the current phase plus $\sin \varphi$ and $\cos \varphi$ as in Eq. 1, and at the start of each episode we set the

joint angles, joint velocities, and forward velocity of the body to the reference values for a uniformly sampled starting phase. This is DeepMimic’s Reference State Initialization. The forward-velocity init matters: without it the body is dropped mid-stride at zero forward speed and immediately falls.

The reward sums four DeepMimic tracking terms (exponential penalties on joint pose, joint velocity, foot end-effector position, and root height) and terminates the episode if forward velocity drops below 0.3 m/s. Our best policy uses a worst-joint pose aggregator (minimum over per-joint penalties) so the policy cannot hide a stiff joint behind compliant ones.

Behavioral cloning warm-start. We pretrain on PD-controller rollouts inside the simulator before PPO fine-tuning. Inverse-dynamics torques would ignore ground reaction forces and be systematically wrong during stance; PD rollouts inside the full simulator produce torques consistent with stance-phase loading.

4.2 Adversarial Motion Priors and AIRL

As a comparison track we implement AMP [7, 2] on top of the same Walker2d environment. A discriminator is trained on state-transition pairs in 12-dimensional joint space using the LSGAN loss with a zero-centered gradient penalty on expert samples. The bounded style reward $r_s = \max(0, 1 - 0.25(D - 1)^2)$ replaces the hand-crafted pose reward, combined with a forward-velocity task reward as $r = w_g r_{\text{task}} + w_s r_s$ with $w_g = 0.35$, $w_s = 0.65$. We additionally implement AIRL [3] with the same discriminator backbone and a state-only reward parameterization.

4.3 Held-out biomechanical-realism scorecard

For each policy we run 6 deterministic rollouts of 2500 steps, detect heel-strikes from per-foot contact-force rising edges with a 0.5 s debounce, and slice into per-stride windows. Per-stride medians are computed for stride period, cadence, double-support fraction, peak vGRF (body-weight units), per-joint ROM, and L-R asymmetry. Reference values use the same methodology on Subject 1’s force-plate and IK data. Each metric contributes to a 0–4 progress score via a $\pm 20\%$ credible band, scaling linearly from 1.0 inside the band to 0 at its edge.

5 Experimental Design

Hypotheses.

- H1.** A phase-conditioned observation with multi-term imitation reward produces *qualitatively* walking policies (balance maintained, bilateral foot alternation, forward translation) where a phase-blind policy with only forward velocity does not.
- H2.** The multi-term reward is exploitable. Each unconstrained degree of freedom yields a characteristic Goodhart-style failure (ankle paddling, one-legged hopping, toe-walking, controlled forward fall, stand-and-wiggle).
- H3.** AMP at fewer than ~ 100 parallel environments produces discriminator collapse: style reward converges to near-zero, eliminating the style gradient.
- H4.** (*Headline.*) Even when H1 holds and the policy visually walks, no engineered-reward variant we tested passes the held-out biomechanical-realism scorecard against force-plate-derived targets.
- H5.** Failure modes that look reward-driven can have structural causes such as joint-range limits in the simulator’s MJCF. Diagnosing reward issues without first checking the kinematic feasibility of the reference is unsafe.

Domain and data. Simulator: MuJoCo 3.3.0 [12] via Gymnasium Walker2d-v4. Reference motion: inverse-kinematics reconstruction of treadmill walking from the SimTK Coordination Retraining dataset [13] (Subject 1, baseline trial, 1.25 m/s), processed into one clean gait cycle of 56 frames at 50 Hz and resampled to 140 frames at 125 Hz via cubic spline. Policy network: 256×256 MLP, tanh activations (Stable-Baselines3 [8] defaults). AMP discriminator: two hidden layers of 64 units, ELU. Training: 8 parallel CPU environments, 5M steps per run, PPO with $n_{\text{steps}} = 2048$, batch size 64, $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, learning rate 3×10^{-4} .

Trained variants. Four post-pivot policies are evaluated against the scorecard: *Baseline* (*stock joints*); *Permissive hips* (-30° to $+60^\circ$ thigh range); *Constrained hips* (-150° to $+35^\circ$, 5° of headroom over the reference’s flexion peak); and *Best engineered* (*lead*), the permissive MJCF plus the worst-joint pose aggregator. AMP/AIRL collapse before producing clean strides and are evaluated on training-time discriminator/style-reward curves (Sec. 6.3).

6 Results and Discussion

6.1 Phase-conditioned PPO produces visually walking policies (H1)

All four variants converge to forward-translating policies within 5M steps (Fig. 1), with episode lengths reaching 3500–5500 steps and returns 1100–3800. Visual inspection of deterministic rollouts confirms balance, foot alternation, and forward progression near target speed. H1 is supported. Notably, the variant that ranks highest on the held-out scorecard ranks *lowest* on training return (*Best engineered*, blue): the worst-joint aggregator is harder to optimize but closes a reward-hacking loophole. Training-time return is not a reliable proxy for biomechanical realism.

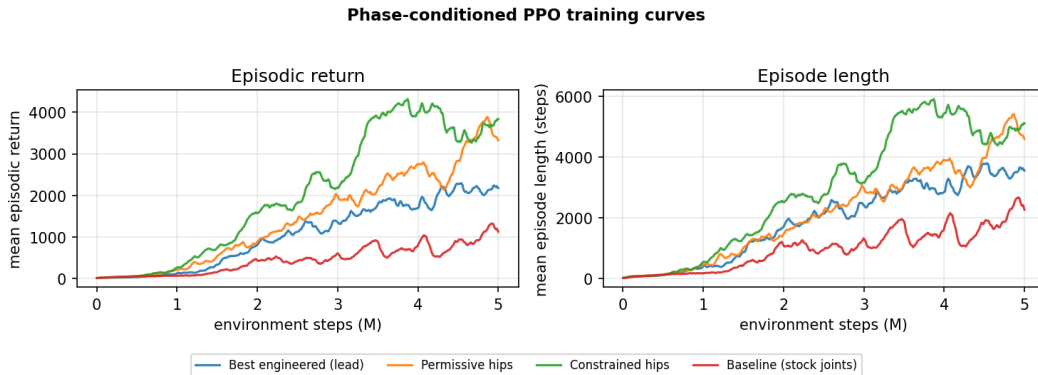


Figure 1: Phase-conditioned PPO training curves over 5M environment steps, EMA-smoothed. The lead policy (blue) by held-out biomech realism ranks last on training-time return. Training-time metrics are not a reliable proxy for what a held-out evaluation will say.

6.2 Reward exploits are structured Goodhart cases (H2)

The multi-term reward arrived through iterative refinement. At each stage, the optimizer found a locally optimal degenerate strategy, and a new term was added to close it. Table 1 catalogues the characteristic exploits, each locally optimal for the partial reward as written. None required pathological optimizer behavior. H2 is supported.

6.3 AMP collapses at small environment count (H3)

We trained AMP and AIRL (including warm-started variants from a working PPO policy) on the same environment. At 8 and 16 parallel environments, the style reward r_s never enters the target band $[0.5, 1.0]$: it starts at 0.22–0.25 and decays toward the collapse floor ~ 0.03 over 5M steps (Fig. 2), with discriminator loss plummeting as separation becomes near-perfect. The mechanism is structural: ~ 140 expert transitions on a 1-D manifold embedded in a 24-D input vs only 8–16 environments of policy diversity — the discriminator has far more capacity than data and memorizes. The 16-env variant collapses more slowly (final $r_s = 0.06$) but does not escape. Escontrela et al. [2] ran 4096 environments on a GPU cluster; the path out is GPU-parallel training (Sec. 7). H3 is supported.

6.4 No engineered-reward variant passes the biomechanical-realism scorecard (H4)

This is the central result of the paper. Table 2 shows per-metric values for the four variants against the measured Subject 1 baseline. Every variant misses every spatiotemporal and kinetic target by $\geq 40\%$. Stride period 0.57–0.67 s vs 1.12 s (cadence 1.7–2.0 $\times$ too fast); peak vGRF 3.7–4.8 $\times$ body weight

Table 1: Reward-hacking failure modes encountered during reward design. Each is a canonical Goodhart case: the policy maximizes the partial reward by exploiting a degree of freedom that was unconstrained by the reward as written. The last row foreshadows the morphology issue analyzed in detail in Sec. 6.5.

Exploit	Mechanism	Closing patch
Ankle paddling	Reference ankle oscillates regardless of foot contact, so rapid ankle motion alone satisfies the joint-tracking reward with no forward progress.	Direct swing-foot contact penalty $-w_{\text{swing}} \tanh(F_{\text{swing}}/50)$.
One-legged hopping	Planting one foot and using the other as a balance pole satisfies stance-leg pose tracking without requiring bilateral coordination.	Explicit contact-alternation reward keyed to the reference stance foot.
Toe-walking	Permanent ankle plantarflexion satisfies the ankle-pose reward by walking on the toes.	End-effector tracking on foot world position.
Controlled forward fall	The agent leans forward to maximize $\dot{x}$ while staying above the height termination threshold until the lean is irrecoverable.	Pitch magnitude included in the root-tracking term and pitch-magnitude termination.
Stand-and-wiggle	A mean-aggregator pose reward is satisfied by wiggling five of six joints while the hip stays stiff, hiding one bad joint behind the average.	Worst-joint aggregator: take the minimum over per-joint penalties instead of the mean.
Stiff-hip stand	With stock MJCF's $[-150^\circ, 0^\circ]$ thigh range, the hip parks at the $+0^\circ$ wall (reference asks for $+30^\circ$) and the policy minimizes the unreachable error by holding still.	Open the MJCF thigh range to admit the reference flexion peak (see Sec. 6.5).

AMP discriminator collapse at small env count

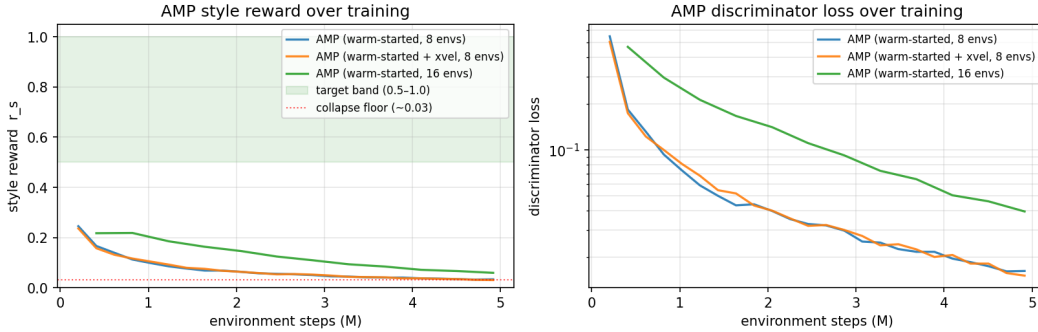


Figure 2: AMP discriminator collapse at small environment count. Style reward (left) never enters the target band (green); 8-env warm-started variants converge to the collapse floor of 0.03, 16-env variants to 0.06. Discriminator loss (right, log scale) drops as the discriminator achieves near-perfect separation, eliminating the style gradient.

vs 1.10; double-support 0.7–7% vs 23%. These are the kinetic signatures of running or bouncing, not walking — walking has $\sim 23\%$ double-support and peak vGRF near body weight, running has near-zero double-support and $\sim 2.5\times$ vGRF. Our values exceed even running.

Figure 3 visualizes the same data as joint-angle overlays, vGRF stance curves, and phase-plane portraits. Kinematic shapes roughly track the reference, but the vGRF curves diverge sharply: where the reference shows the classic double-hump walking trace near $1\times$ body weight, the policies produce spikes of $3.7\text{--}4.8\times$. The hip-knee phase planes show much smaller cycles than the reference, the signature of reduced ROM and shorter stride. H4 is confirmed.

Discussion. The torque actuator has no mechanism by which producing 4 to 5 times body-weight ground reaction forces is more expensive than producing $1\times$ — both are just torque commands. Closing the gap inside this paradigm would require explicit reward terms on double-support, vGRF

Walker2d biomechanical realism — sim policies vs Ulrich Subject 1 baseline

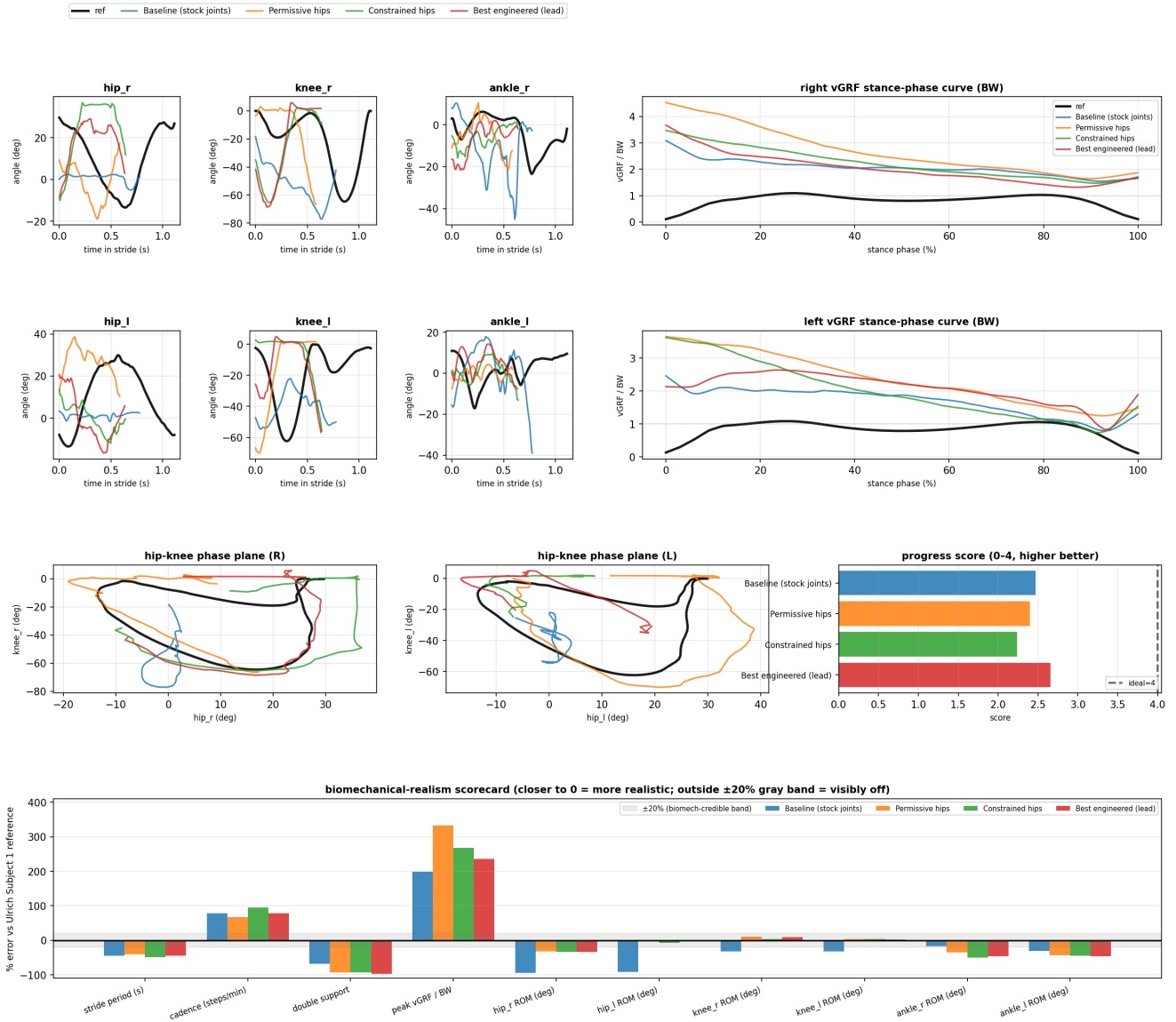


Figure 3: Biomechanical-realism dashboard. Top two rows: per-joint angle traces over one stride, sim variants overlaid on the measured reference (black). Right: vertical ground reaction force during stance, sim vs reference. Third row: hip-knee phase planes (R, L) and progress score. Bottom: $\pm 20\%$ credible-band scorecard, closer to zero is more realistic. No sim variant lies inside the band on any spatiotemporal or kinetic metric.

Table 2: Held-out biomechanical-realism scorecard. Sim values are per-stride medians over 6 deterministic episodes $\times$ 2500 steps; reference values are per-stride medians from Subject 1’s own force-plate and IK data. Percent error vs reference in parentheses. No variant passes the $\pm 20\%$ credible band on any metric except mirror-leg knee ROM.

metric	reference	Baseline (stock)	Permissive hips	Constrained hips	Best (lead) engineered
stride period (s)	1.12	0.63 (-44%)	0.67 (-40%)	0.57 (-49%)	0.63 (-44%)
cadence (steps/min)	107	191 (+78%)	179 (+67%)	210 (+96%)	192 (+79%)
double-support fraction	0.227	0.072 (-68%)	0.017 (-92%)	0.017 (-92%)	0.007 (-97%)
peak vGRF / BW	1.10	3.29 (+199%)	4.77 (+333%)	4.05 (+268%)	3.70 (+236%)
hip ROM (deg, R)	45.4	2.78 (-94%)	30.6 (-33%)	30.5 (-33%)	30.2 (-34%)
knee ROM (deg, R)	65.7	44.4 (-32%)	72.7 (+11%)	68.2 (+4%)	72.2 (+10%)
ankle ROM (deg, R)	40.0	33.4 (-17%)	26.4 (-34%)	19.8 (-51%)	21.8 (-45%)
progress score (0–4)	4.00	2.47	2.40	2.24	2.66

magnitude, and stride period, which is no longer imitation but hand-engineered gait shaping with the imitation cost as a regularizer. We read the negative result as structural to the engineered-reward + torque-actuated combination, not as a tuning failure.

6.5 Morphology can mask reward problems (H5)

Earlier batches chased a “stiff-hip” failure (trained hip range $\sim 2^\circ$ vs a reference asking for $\sim 45^\circ$). The root cause was not the reward: stock walker2d.xml constrains the thigh to $[-150^\circ, 0^\circ]$, while the reference’s flexion peak is $+30^\circ$. About 68% of every reference cycle lay outside the joint range, and the policy was pinned at the 0° wall for 95% of frames (Fig. 4). Opening the range produced an order-of-magnitude jump in hip ROM. Methodological lesson: check kinematic feasibility before tuning the reward.

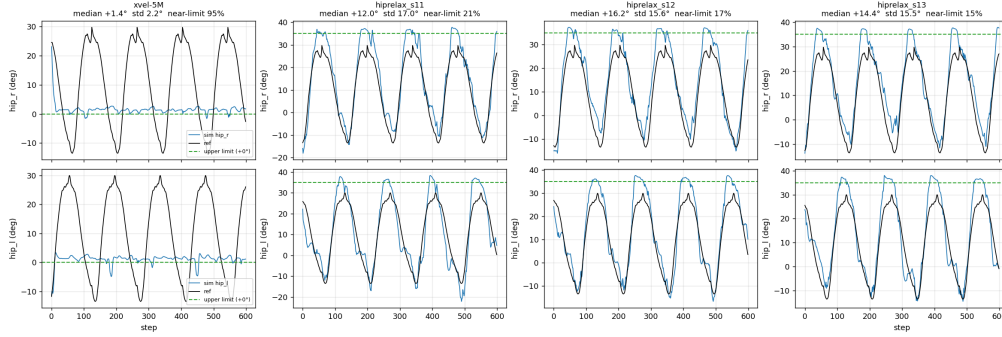


Figure 4: Hip-angle traces. Baseline (left) vs three relaxed-MJCF seeds. The baseline’s hip is parked at the $+0^\circ$ wall (dotted green line); relaxing the joint range lets the policy sweep through the reference range. H5 supported.

7 Conclusion and Future Work

Can engineered-reward phase-conditioned PPO on torque-actuated Walker2d recover a biomechanically realistic gait against held-out force-plate ground truth? The answer is **no**. The visually walking policies produce running-gait kinetics (peak vGRF 3.7–4.8 BW, near-zero double-support); AMP at 8–16 environments collapses before producing clean strides. The engineered reward closes individual reward-hacking loopholes through hand-engineering, but those loopholes do not include peak vGRF or double-support, and no variant crosses the $\pm 20\%$ credible band on those metrics. The torque actuator does not penalize 4-BW foot-slamming, and a reward shape that does penalize it amounts to engineering walking by hand.

Two paths forward. First, port the environment to MuJoCo MJX [4] and train AMP at 2,000–4,000 parallel environments on a single GPU, matching Escontrela et al. [2]. At that scale, policy diversity forces the discriminator to generalize, and r_s should escape the collapse floor. Second, return to the original proposal’s muscle-actuated humanoid scope: KINESIS [11] and KinTwin [1] demonstrate that muscle-driven imitation reproduces able-bodied and impaired gait, including from markerless mocap. Hill-type muscles cannot produce $4\times$ BW peak vGRF without violating force-length-velocity limits — the muscle actuator is, in effect, the implicit biomechanical-realism prior we tried to construct as a reward. Either path, evaluated against the same scorecard, would directly measure the realism gap this paper documents.

Code

https://github.com/Brock-Perreca/6955_Project/tree/main

References

- [1] R. James Cotton. KinTwin: Imitation learning with torque and muscle driven biomechanical models enables precise replication of able-bodied and impaired movement from markerless motion capture. *arXiv preprint arXiv:2505.13436*, 2025.
- [2] Alejandro Escontrela, Xue Bin Peng, Wenhao Yu, Tingnan Zhang, Atil Iscen, Ken Goldberg, and Pieter Abbeel. Adversarial motion priors make good substitutes for complex reward functions. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.
- [3] Justin Fu, Katie Luo, and Sergey Levine. Learning robust rewards with adversarial inverse reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2018.
- [4] Google DeepMind. MuJoCo MJX: A JAX-based reimplement of MuJoCo. https://github.com/google-deepmind/mujoco_mjx, 2024.
- [5] Jonathan Ho and Stefano Ermon. Generative adversarial imitation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- [6] Xue Bin Peng, Pieter Abbeel, Sergey Levine, and Michiel van de Panne. DeepMimic: Example-guided deep reinforcement learning of physics-based character skills. *ACM Transactions on Graphics (SIGGRAPH)*, 37(4), 2018.
- [7] Xue Bin Peng, Ze Ma, Pieter Abbeel, Sergey Levine, and Angjoo Kanazawa. AMP: Adversarial motion priors for stylized physics-based character control. *ACM Transactions on Graphics (SIGGRAPH)*, 40(4), 2021.
- [8] Antonin Raffin, Ashley Hill, Adam Gleave, Anssi Kanervisto, Maximilian Ernestus, and Noah Dormann. Stable-baselines3: Reliable reinforcement learning implementations. *Journal of Machine Learning Research*, 22(268), 2021.
- [9] Nikita Rudin, David Hoeller, Philipp Reist, and Marco Hutter. Learning to walk in minutes using massively parallel deep reinforcement learning. In *Conference on Robot Learning (CoRL)*, 2022.
- [10] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- [11] Merkourios Simos, Antoine Chiappa, and Alexander Mathis. KINESIS: Reinforcement learning-based motion imitation for physiologically plausible musculoskeletal motor control. *arXiv preprint arXiv:2503.14637*, 2025.
- [12] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012.
- [13] Scott D. Uhlich and Scott L. Delp. Muscle coordination retraining to reduce knee loading: Treadmill walking dataset. SimTK project repository, 2018. <https://simtk.org/projects/coordretraining>.

- [14] Scott D. Uhlich, Antoine Falisse, Łukasz Kidziński, Julie Muccini, Michael Ko, Akshay S. Chaudhari, Jennifer L. Hicks, and Scott L. Delp. OpenCap: Human movement dynamics from smartphone videos. *PLOS Computational Biology*, 19(10):e1011462, 2023.